PROBLEM DA7

Thrust of a Submarine Shrouded Fan

Original statement · DA7.pdf

Π theorem | Fractional exponents | Re ≫ 1

OFFICIAL PROBLEM STATEMENT

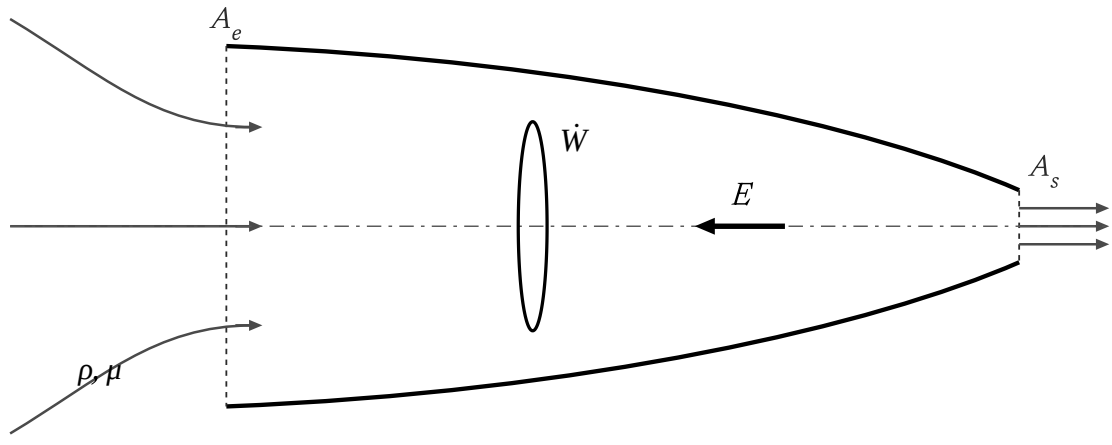
The submarine propulsion device schematically represented in the figure consists of a nozzle, with inlet and outlet crosssection area A_e and A_s, and a fan located inside, which imparts a rate of work per unit time (power) $\dot{W}$ to the circulating water stream, whose density and viscosity are ρ and μ, respectively. To characterize the thrust E produced when the submarine is at rest and at sufficient depth for gravity to have a negligible effect, please follow these steps:

1. By application of the Π theorem, reduce the parametric dependence of the thrust E.

2. Simplify the result assuming that the viscosity has a negligible effect, **justifying** the simplification. Maintain this assumption in addressing the next question.

3. An experimental test is conducted with a geometrically similar model with scale 1/10. It is found that, for a fan power $\dot{W}_e = 100$ Watts the resulting thrust is $E_e = 125$ N. If a thrust $E_p = 12500$ N is needed for the real prototype, determine the power W necessary to drive the fan.

The subscript of the prototype thrust is blurred in the scan; it reads as E_p (prototype).



nozzle with a fan of power Ẇ; thrust E with the submarine at rest

Statement figure (redrawn from DA7.pdf): a nozzle of inlet area A_e and outlet area A_s with a fan inside that gives the power Ẇ to water of density ρ and viscosity μ; E is the thrust

Source: DA7.pdf (page 1, exam of 17-05-2010, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; limits, Eq. 6.55; physical similarity, Eqs. 6.57-6.59.

Worked solution

Hypotheses, Data and Dimensional Table

- The submarine is at rest, so there is no free-stream velocity: the only "motor" of the flow is the power $\dot{W}$. Gravity is negligible (deep water, no free surface).
- The nozzle is defined by its two areas (the rest of the shape by geometric similarity). Same water in the model and in the prototype.

Parameter	Meaning	Dimensions
E	thrust (unknown)	MLT^{-2}
$\dot{W}$	fan power	ML^2T^{-3}
ρ	water density	ML^{-3}
μ	water viscosity	$ML^{-1}T^{-1}$
A_e, A_s	inlet and outlet areas	L^2

Application of the Π Theorem

$$E = E(\dot{W}, \rho, \mu, A_e, A_s)$$

$n = 5$. Since no velocity is given, the independent set must use the power: $(\rho, \dot{W}, A_s)$. Rows in (L, M, T): $\rho(-3, 1, 0)$, $\dot{W}(2, 1, -3)$, $A_s(2, 0, 0)$. Expanding along the third row, $\det = 2 \cdot (1 \cdot (-3) - 0 \cdot 1) = -6 \neq 0$, so $k = 3$ and two groups remain.

Thrust. $[E] = [\rho]^a [\dot{W}]^b [A_s]^c$:

$$M:\ 1 = a + b, \qquad T:\ -2 = -3b, \qquad L:\ 1 = -3a + 2b + 2c$$

so $b = 2/3$, $a = 1/3$ and $2c = 1 + 1 - \frac{4}{3} = \frac{2}{3}$, $c = 1/3$: the scale $\rho^{1/3} \dot{W}^{2/3} A_s^{1/3}$.

Viscosity. $[\mu] = [\rho]^a [\dot{W}]^b [A_s]^c$:

$$M:\ 1=a+b, \qquad T:\ -1=-3b, \qquad L:\ -1=-3a+2b+2c$$

so $b=1/3$, $a=2/3$ and $2c=-1+2-\frac{2}{3}=\frac{1}{3}$, $c=1/6$. The group is $\mu/(\rho^{2/3}\dot{W}^{1/3}A_s^{1/6})$; its inverse plays the role of a Reynolds number.

Areas. A_e/A_s.

$$\frac{E}{\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}}=f\left(\frac{A_e}{A_s},\frac{\rho^{2/3}A_s^{1/6}\dot{W}^{1/3}}{\mu}\right)$$

Check that the last group is a Reynolds number: a velocity scale built from the power is $U_s=(\dot{W}/(\rho A_s))^{1/3}$ (kinetic-energy flux $\rho U^3 A\sim\dot{W}$), and $\rho U_s A_s^{1/2}/\mu=\rho^{2/3}\dot{W}^{1/3}A_s^{1/2-1/3}/\mu=\rho^{2/3}\dot{W}^{1/3}A_s^{1/6}/\mu$, the same group.

Negligible Viscosity

If

$$\frac{\rho^{2/3}A_s^{1/6}\dot{W}^{1/3}}{\mu}\gg 1$$

viscosity is negligible and the group drops out (Notes.pdf, Eq. 6.55):

$$\frac{E}{\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}}=f\left(\frac{A_e}{A_s}\right)$$

Justification with the test data (water, $\rho=10^3$ kg/m^3, $\mu=10^{-3}$ kg/(m s)): the model area is not given, but any reasonable outlet, e.g. $A_s\sim 10^{-2}$ m^2, with $\dot{W}=100$ W gives $\rho^{2/3}A_s^{1/6}\dot{W}^{1/3}/\mu=100\times 0.46\times 4.64/10^{-3}\simeq 2\times 10^5\gg 1$ (an estimate, since the size is assumed).

Physical Similarity and Prototype Power

Geometric similarity gives the same A_e/A_s, so the dimensionless thrust is the same; with the same water:

$$\frac{E_p}{\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}} = \frac{E_e}{\rho^{1/3}\dot{W}_e^{2/3}A_{se}^{1/3}} \implies \dot{W}^{2/3} = \frac{E_p}{E_e}\left(\frac{A_{se}}{A_s}\right)^{1/3}\dot{W}_e^{2/3}$$

and raising to $3/2$:

$$\dot{W} = \left(\frac{E_p}{E_e}\right)^{3/2}\left(\frac{A_{se}}{A_s}\right)^{1/2}\dot{W}_e$$

Scale 1/10 in length means $A_{se}/A_s = 1/100$:

$$\dot{W} = \left(\frac{12500}{125}\right)^{3/2}\left(\frac{1}{100}\right)^{1/2} \times 100\ \text{W} = 100^{3/2} \times \frac{1}{10} \times 100 = 1000 \times \frac{1}{10} \times 100 = 10^4\ \text{W}$$

Answers and Checks

- Reduced law: $E/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = f\left(A_e/A_s,\ \rho^{2/3}A_s^{1/6}\dot{W}^{1/3}/\mu\right)$.
- With $\rho^{2/3}A_s^{1/6}\dot{W}^{1/3}/\mu \gg 1$: $E/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = f(A_e/A_s)$.
- Prototype power: $\dot{W} = 10^4$ W = 10 kW.
- Dimensions: $\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}$ is $M^{1/3}L^{-1} \cdot M^{2/3}L^{4/3}T^{-2} \cdot L^{2/3} = MLT^{-2}$, a force.
- Physical sense: at fixed geometry $E \propto \dot{W}^{2/3}$; a hundred times the thrust with a hundred times the area needs $(100)^{3/2}/(100)^{1/2} = 100$ times the power.
- Check against the official key: DA7.pdf gives $\dot{W} = (E/E_e)^{3/2}(A_{se}/A_s)^{1/2}\dot{W}_e = 10^4$ W, the same result.